

Assignment – 21.2

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Batch: 21

Task 1 – AI-Assisted Pair Programming

- Use AI to generate an initial program (e.g., a simple To-Do List Manager).
- Ask AI to suggest enhancements (such as error handling, additional features, or refactoring).
- Compare the original and improved versions to understand the role of AI in collaboration.

Prompt:

Create a simple To-Do List Manager in Python where users can:

- Add tasks
- View tasks
- Remove tasks

```
lab 19 > 12-04-2026.py > ...
1  #Create a simple To-Do List Manager in Python where users can:
2  #- Add tasks
3  #- View tasks
4  #- Remove tasks
5  class ToDoListManager:
6      def __init__(self):
7          self.tasks = []
8
9      def add_task(self, task):
10         self.tasks.append(task)
11         print(f'Task "{task}" added to the list.')
12
13     def view_tasks(self):
14         if not self.tasks:
15             print("No tasks in the list.")
16         else:
17             print("To-Do List:")
18             for idx, task in enumerate(self.tasks, start=1):
19                 print(f"{idx}. {task}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● To-Do List:
Task "Call Alice" added to the list.
Task "Call Alice" added to the list.
To-Do List:
1. Buy groceries
2. Call Alice
Task "Buy groceries" removed from the list.

○ To-Do List:
1. Call Alice

powerhell
Python
You have Docker Desktop installed. You can install the recommended Docker extension for VS Code.

Explanation

- AI was used to generate the **initial version** of the program
- The first version had **basic functionality only**
- Then AI was used again to:
 - Improve code quality
 - Add error handling (try-except)
 - Validate user input

- Enhance user interface

This shows AI acting as a **pair programmer** by:

- Writing code
- Reviewing code
- Improving code

Task 2 – Version Control Integration

Initialize a Git repository for your project.

Perform the following steps:

- Create and switch to a new branch.
- Add AI-assisted modifications.
- Commit with a meaningful message.
- Merge changes back into the main branch.

Reflect on how version control supports collaborative development with AI assistance.

Prompt:

Explain how to use Git for managing AI-generated code, including:

- Repository initialization
- Branching
- Committing changes
- Merging branches

```
PS C:\Users\dell\OneDrive\Desktop\AI> git add todo.py
PS C:\Users\dell\OneDrive\Desktop\AI> git commit -m "Initial version of To-Do List Manager"
[master (root-commit) ddf3a35] Initial version of To-Do List Manager
1 file changed, 50 insertions(+)
create mode 100644 todo.py
PS C:\Users\dell\OneDrive\Desktop\AI> git branch feature-enhancement
PS C:\Users\dell\OneDrive\Desktop\AI> git checkout feature-enhancement
Switched to branch 'feature-enhancement'
PS C:\Users\dell\OneDrive\Desktop\AI> notepad todo.py
PS C:\Users\dell\OneDrive\Desktop\AI> git add todo.py
PS C:\Users\dell\OneDrive\Desktop\AI> git commit -m "Added welcome message using AI suggestion"
[feature-enhancement 32cca26] Added welcome message using AI suggestion
1 file changed, 1 insertion(+)
PS C:\Users\dell\OneDrive\Desktop\AI> git checkout master
Switched to branch 'master'
PS C:\Users\dell\OneDrive\Desktop\AI> git merge feature-enhancement
Updating ddf3a35..32cca26
Fast-forward
 todo.py | 1 +
1 file changed, 1 insertion(+)
PS C:\Users\dell\OneDrive\Desktop\AI>
```

Explanation

- A Git repository was initialized using git init
- The To-Do List file was added and committed
- A new branch feature-enhancement was created
- AI-based improvements were added in the branch
- Changes were merged back into the main branch

Git helps in:

- Tracking changes
- Managing multiple versions
- Supporting collaboration
- Safely merging improvements

Task 3 – AI for Commit Message Generation

- Use AI to generate concise, meaningful commit messages

based on the changes made.

- Compare AI-generated messages with human-written ones.
- Discuss how AI can improve documentation and traceability in Git history.

Prompt:

Generate a concise and meaningful commit message for adding a welcome message and improving user interaction in a To-Do List application

```
print( "welcome to AI To-Do Manager" )
tasks = []

def show_tasks():
    if not tasks:
        print("No tasks available.")
    else:
        print("\nTo-Do List:")
        for i, task in enumerate(tasks, 1):
            print(f"{i}. {task}")

while True:
    print("\n==== TO-DO LIST MENU =====")
    print("1. Add Task")
    print("2. View Tasks")
    print("3. Remove Task")
    print("4. Exit")

    try:
        choice = int(input("Enter your choice: "))

        if choice == 1:
            task = input("Enter task: ")
            if task.strip():
```

Explanation:

Explanation

- AI generated a **clear and concise commit message**
- Compared with human-written message:
 - AI → more structured and consistent
 - Human → more specific but less uniform

AI improves:

- Documentation
- Readability
- Traceability in Git history

Task 4 – Pair Programming Simulation

- Student A writes the initial version of a program.
- Student B, with AI assistance, suggests improvements or adds features.
- Both students collaborate using Git branching and merging to integrate their work.

Prompt:

Simulate pair programming where:

- One student writes the initial version of a program
- Another student uses AI to improve it
- Use Git branching and merging to combine work

```
lab 19 > 12-04-2026.py > ...
1 #Generate a concise and meaningful commit message for adding a we
2 #Add welcome message and enhance user interaction in To-Do List app
3 #- Implemented a welcome message to greet users upon launching the appl
4 #- Improved user interaction by adding prompts and feedback for better
5 #- Enhanced the overall user experience with a more engaging inter face.
6 # Welcome message for the To-Do List application
7 def welcome_message():
8     print("Welcome to the To-Do List Application!")
9     print("Manage your tasks efficiently and stay organized.")
10
11 # Call the welcome message when the application starts
12 welcome_message()
13
```

esktop/AI/lab 19/12-04-2026.py"

- Welcome to the To-Do List Application!
- Manage your tasks efficiently and stay organized.

PS C:\Users\dell\OneDrive\Desktop\AI>

Explanation

- Student A created the **basic To-Do application**
- Student B (with AI):
 - Added new features
 - Improved code quality
- Git was used to:
 - Create a separate branch
 - Add improvements
 - Merge changes into main branch

This demonstrates:

- Real-world collaboration
- AI-assisted development
- Efficient teamwork using Git

